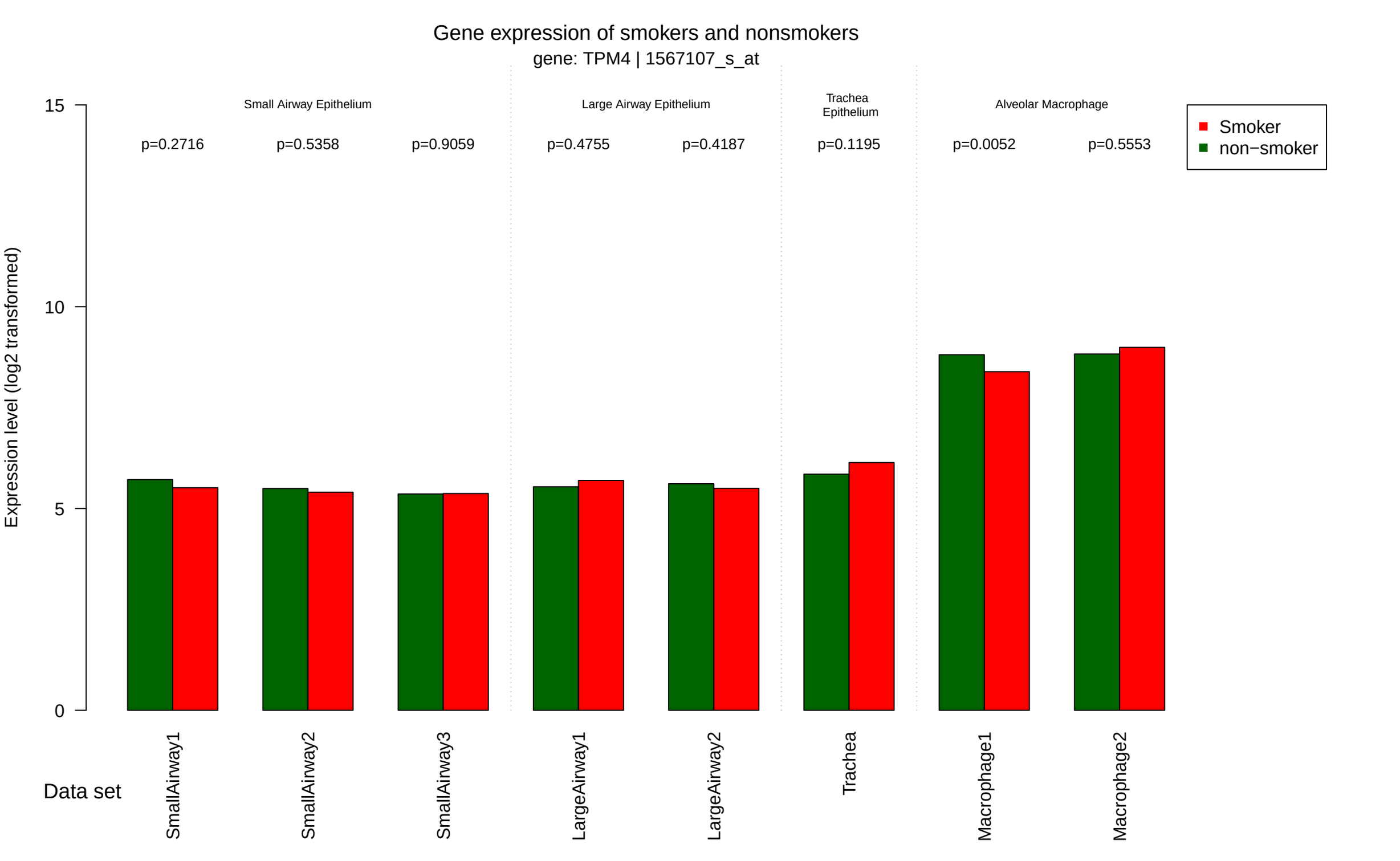
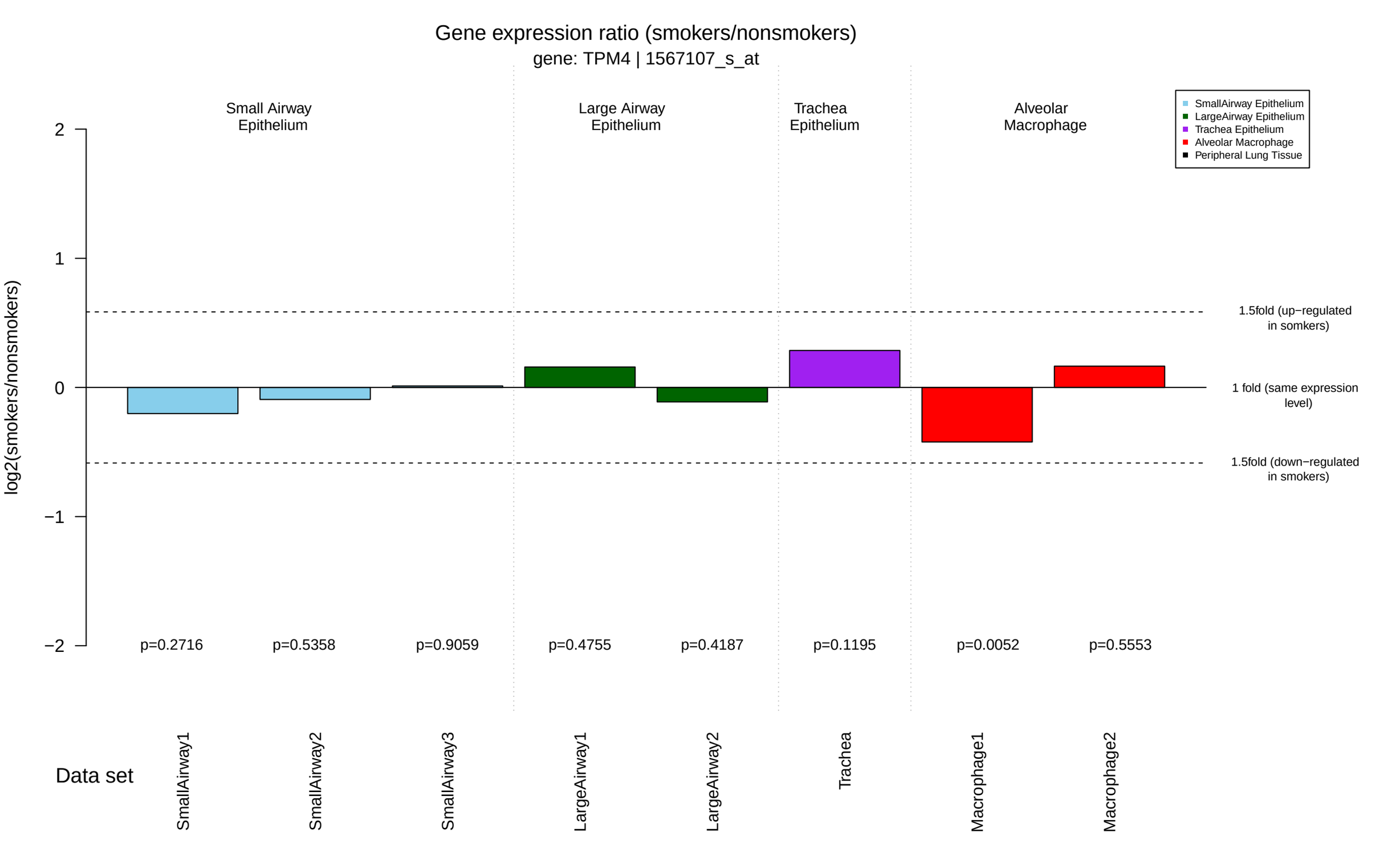
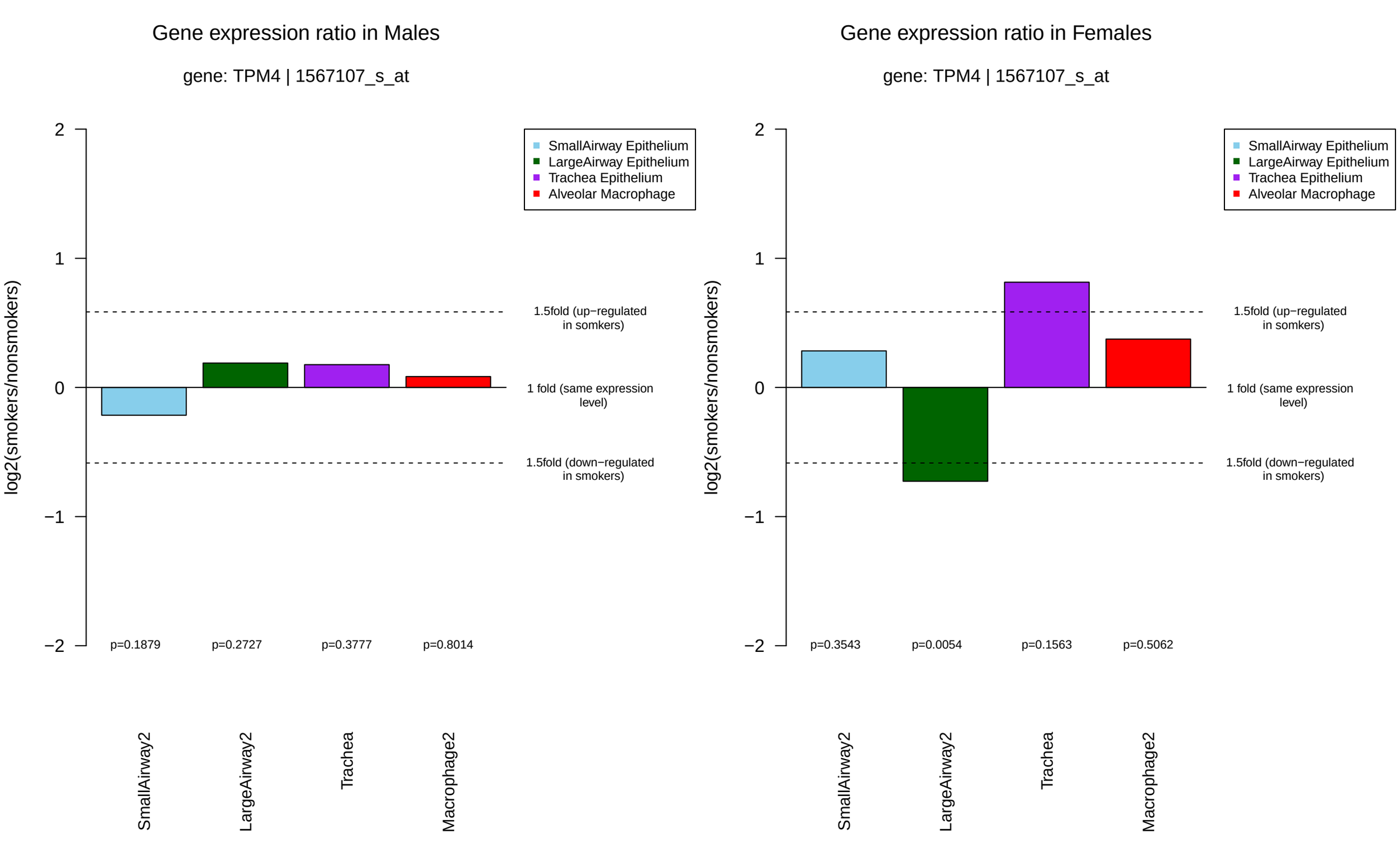
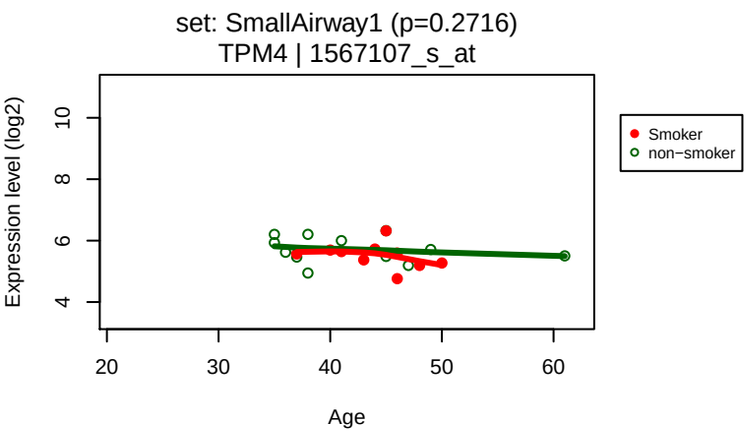
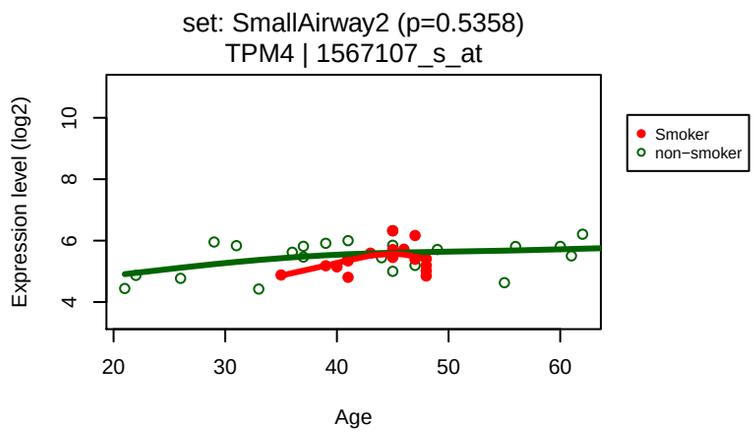
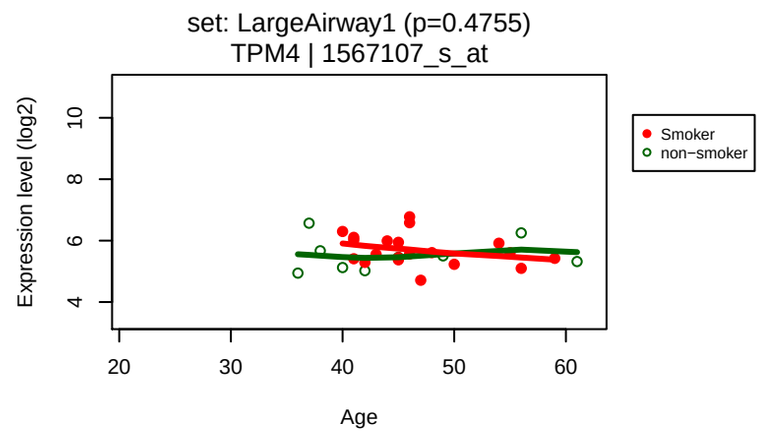

Gene expression of smokers and nonsmokers
gene: TPM4 | 1567107_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2716
p=0.5358
p=0.9059
p=0.4755
p=0.4187
p=0.1195
p=0.0052
p=0.5553
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

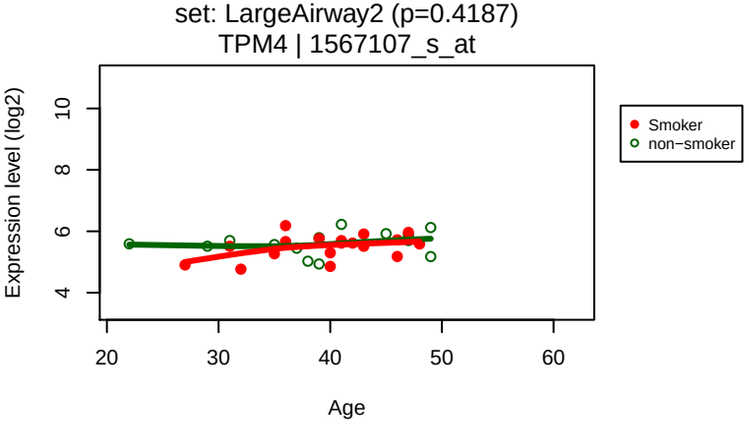

Gene expression ratio (smokers/nonsmokers)
gene: TPM4 | 1567107_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2716
p=0.5358
p=0.9059
p=0.4755
p=0.4187
p=0.1195
p=0.0052
p=0.5553
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

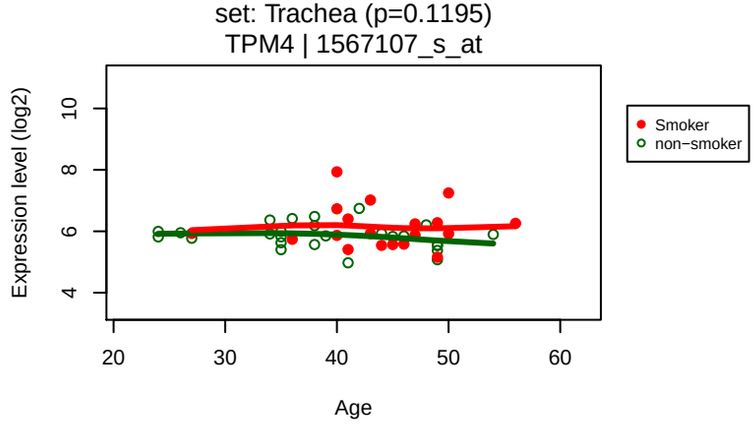

Gene expression ratio in Males
gene: TPM4 | 1567107_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1879
p=0.2727
p=0.3777
p=0.8014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TPM4 | 1567107_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3543
p=0.0054
p=0.1563
p=0.5062
SmallAirway2
LargeAirway2
Trachea
Macrophage2

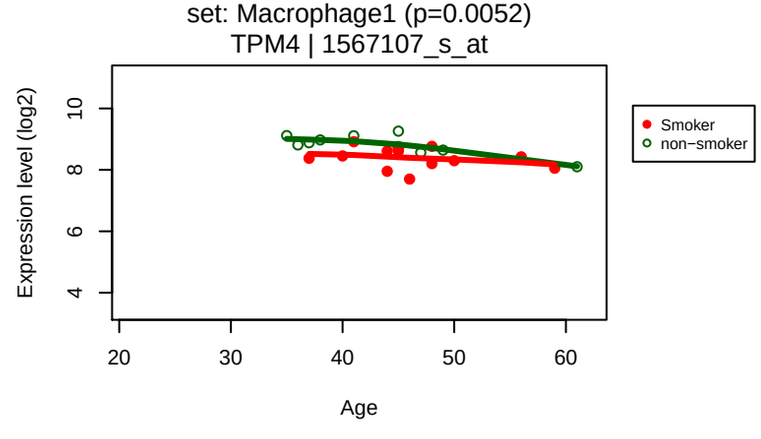

set: SmallAirway1 (p=0.2716)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker

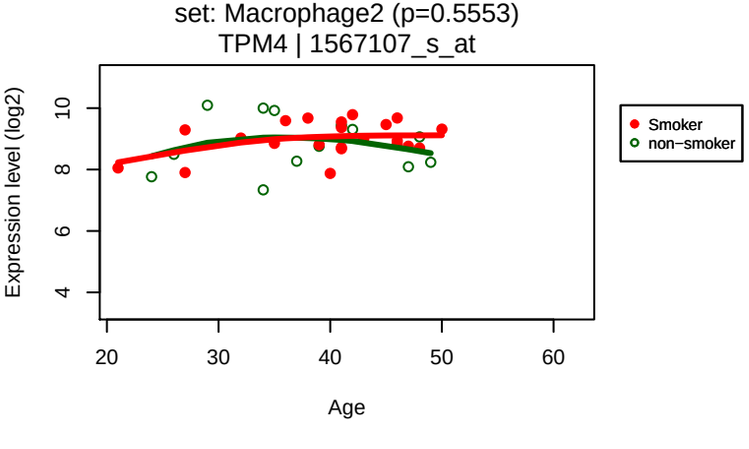

set: SmallAirway2 (p=0.5358)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker

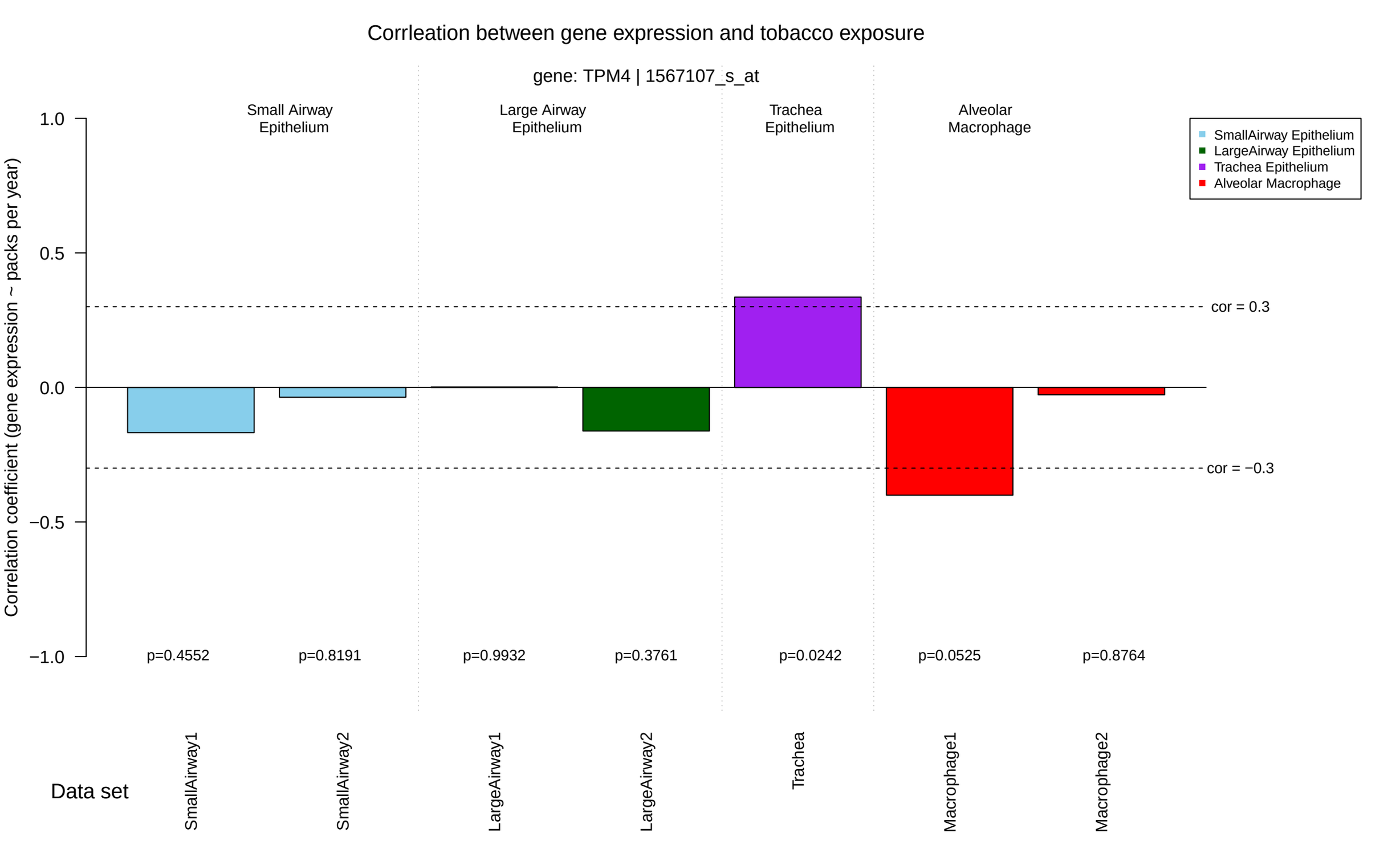

set: LargeAirway1 (p=0.4755)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4187)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1195)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0052)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5553)
TPM4 | 1567107_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TPM4 | 1567107_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4552
p=0.8191
p=0.9932
p=0.3761
p=0.0242
p=0.0525
p=0.8764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2